

Clinical Risk Assessment Report: Patient 28

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 10.2% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.46x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.5%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.31 [Avg (32th %ile)] (-2.5%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: -0.47 [Avg (29th %ile)] (-1.2%)**
Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 0.08 [Avg (48th %ile)] (-0.8%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

Tumor Local Contrast (PET) (GLCM_Contrast(=Variance).PET) **Value: -0.64 [Bottom 19%] (-0.8%)**
Measures local variations and sharp transitions in density or metabolism, signifying an unstructured and variable tumor matrix.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.29 [Avg (50th %ile)] (-0.6%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Tumor Sphericity (PET) (SHAPE_Sphericity(onlyFor3DROI)).PET) **Value: -0.89 [Bottom 16%] (-0.4%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Patient Age (Age) **Value: 61.0 [Avg (40th %ile)] (-0.4%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.39 [Bottom <1%] (-0.4%)**
Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Machine Learning Feature Attribution (SHAP Decision Path)

